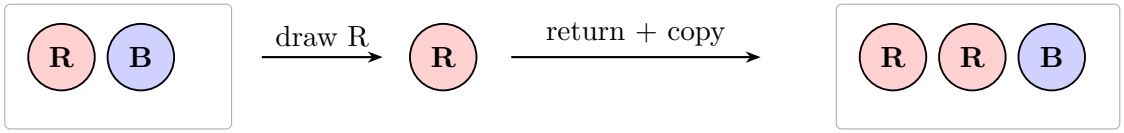
Start with one red and one blue. Draw, return it, add one counter of that color, and mix. For random play, each counter has the same chance; for possible stories, choose the draws yourself.



Marks distinguish counters of the same color. Start with R1 and B1. The first added counter gets mark 2; the next gets mark 3.



draw R1 return it and add R2 next draw could be R2

Problem 1: Find all the two-draw marked histories. Use them to decide whether zero, one, or two red draws has the greatest chance, if any.

Problem 2: Keep the starting bag R1, B1, but return each draw without adding a copy. How does the chance of zero, one, or two red draws change? Explain the difference exactly.

Problem 3: Return to copying. Compare the color stories RRB, RBR, and BRR. Does their order change their chance? Give an exact reason for your answer.

RRB

RBR

BRR

Problem 4: Three copying draws have 24 equally likely marked histories. How many give zero, one, two, or three red draws? Explain how you know your counts cover every possibility.

0 red draws

1 red draw

2 red draws

3 red draws

Problem 5: Predict the chances of zero, one, two, three, or four red draws after four copying draws. Check your prediction exactly without making a 120-row list.

Problem 6: What do you predict about red-draw totals after five draws, or after more draws? Explore your prediction.